7 **(a) (i)** A field of force is a region of space where a body, by virtue of the property it possesses, experiences a force.

(ii) The electric field strength at a point in an electric field is defined as the electrostatic force per unit positive charge, exerted on the positive test charge placed at that point.

(iii) If the test charge is moving, there is a possibility that the force measured may be due to its motion in a magnetic field, e.g. Earth's magnetic field is always present. This will cause inaccuracies in the measurement of the actual electric field strength.

(b) (i) The field strength of the field at a point is numerically equal to the potential gradient at that point and is given by $E = -\frac{dV}{dx}$.

(ii) 1. Since the direction of the electric field E is in the direction of decreasing electric potential, at point P, it is towards sphere B.

2. At x = 35.0 cm, the potential gradient at that point is zero. Thus, since field strength of the field at a point is numerically equal to the potential gradient at that point, the field strength is zero and force on electron is also zero

3. Inside a charged conductor at equilibrium, there is no further movement of charges as no electric force is acting on the charges. This implies that the electric field strength is zero. x = 0 to x = 9.0 cm is inside Sphere A and thus the potential gradient is zero, implying a constant potential.

(c) (i) Choose any 2 points to test the relationship.
At x = 11 cm, V = 680 V, Vx = 7480
At x = 21 cm, V = 390 V, Vx = 8190

Since the 2 values are not equal, the expression Vx = constant is incorrect.

(ii) The relationship Vx = constant is applicable only to isolated point charge where $V \propto 1/r$. However, the two charged objects present are not point charges and thus the relationship is not valid.

(d) (i) The electron is attracted by the charged spheres and for it to have minimum speed, it will cross the line joining the centres of the two spheres at the point of minimum potential. This is so that the loss in electric potential energy 300e J is minimum.
By conservation of energy, loss in E.P.E = gain in K.E.

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$$eV = \frac{1}{2}mv^2 \Rightarrow \text{minimum speed, } v = \sqrt{\frac{2eV}{m}}$$

$$= \sqrt{\frac{2(1.6\times10^{-19})(295)}{9.11\times10^{-31}}}$$

$$= 1.02\times10^{7} \text{ m s}^{-1}$$

(ii) The electron is first attracted by either one of the positively charged spheres. If the electron is a first moving to the left of the minimum potential point, there will be an electric force on it to the right and vice versa. This electric force present will cause the electron to move left and right as it approaches the spheres till it crosses the line at $x = 0.350$ m.